Modelling motion with graphs

STUDENT INFORMATION AND EXAMPLES

Reading a graph

A motion graph shows how a measured quantity changes with time; it is NOT a picture of the car's route. Always put time (s) on the horizontal axis and position (m) or velocity (m/s) on the vertical axis.

Plot measured points with appropriate scales, labelled axes, units and a suitable straight line or smooth curve of best fit. Do not join noisy experimental points with sharp zigzags.

A negative position or velocity is a signed value in the selected coordinate direction; negative does not mean physically impossible.

Position–time graphs

Gradient (slope) = change in vertical value ÷ change in horizontal value = $\Delta x/\Delta t$. Its unit is m/s, so it represents velocity.

- Positive slope: velocity is in the positive direction. Negative slope: velocity is in the negative direction. Zero slope: velocity is zero.
- Steeper position–time slope means greater speed; a curved position–time graph shows changing velocity.

For a straight line, calculate the slope using two widely spaced points on the line. For a curve, draw a tangent at the time of interest and calculate the tangent gradient.

Velocity–time graphs

Gradient = $\Delta v/\Delta t$. Its SI unit is (m/s)/s = m/s^2, so it represents acceleration.

- Horizontal v–t graph: constant velocity and zero acceleration, even when v is not zero.
- Positive gradient: positive acceleration; negative gradient: negative acceleration.
- A graph may cross v = 0 when the car changes direction; graph position and motion direction carefully.

Extension: the signed area between a velocity–time curve and the time axis gives displacement. This is a secondary idea here.

Graphing and motion-sensor activity

Follow the teacher's motion-sensor challenges. Discuss where the detector takes x = 0, which direction is positive, and how changing speed alters the trace.

Suggested online follow-up: https://glidepathscience.org/simulations/motion-lab/ or https://www.walter-fendt.de/html5/phen/ (choose a suitable motion simulation).